

Electricity Markets

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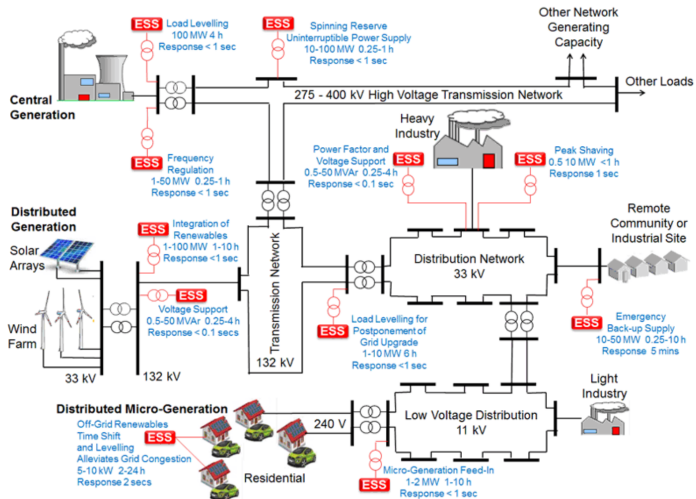
University of Alberta

February 23, 2025

Why aspiring lawyers should care about electricity and what you need to know

- Regulatory complexity:
 - Complex regulation, even in *deregulated* jurisdictions
 - Each part of the market has its own set of regulatory constructs (distribution, transmission, generation, retail)
- New technology means evolving market structures:
 - Electricity is, arguably, changing faster than any other energy market
 - Alberta's electricity market in the midst of a period of market- and regulatory-driven transition
- Economics 101 in action
 - Supply and demand curves determine prices every hour in Alberta's power market
- The next constitutional battleground?
 - Fed/prov issues, environmental policy, section 125 jurisprudence, the oddity that is section 92(10) including the declaratory power, and POGG all come into play!

Grid Energy Storage Systems (ESS) and Applications



Source: http://www.mpoweruk.com/grid_storage.htm

Market Participants

- Generation
- Transmission
- Distribution
- Ancillary Services
- Load
- Storage
- Microgeneration

Market Regulation in Alberta

- Generation is a competitive market
- Transmission is regulated on a cost-of-service basis
- Distribution (local wires) is regulated on a cost-of-service basis, but retail (billing) is competitive.
- Ancillary Services is competitive
- Load (i.e. customers) may contract for electricity supply
- Storage (Still a lot TBD)

Energy units - electricity

- Watts: measure of capacity (instantaneous production, installed capacity, or instantaneous demand)
 - Alberta system demand: 7,200-10,700 MW (megawatts or million watts)
 - Capital Power's Genessee 3 power plant has a nameplate capacity of 450 MW
- Watt hours: measure of energy (production or demand during a given period of time; i.e. flow through)
 - Production over a day, week, month, year
 - A 7.5W LED bulb on all day (24hr) uses 180Wh of electricity (.18kWh)
- Volts: measure of the electrical potential or the ability to convert charge to power (Watts=amps x volts)
 - Transmission lines: 150-765 kV
 - Distribution lines: 13,800 Volts
 - Household wiring: 120-240 Volts

Energy Prices

- Electricity prices: expressed in power delivered over time
 - Cents/kilowatt-hour (c/kWh)
 - Dollars per megawatt-hour (\$/MWh)
 - Levelized costs of electricity (supply costs) in \$/MWh
- Capacity costs are expressed in a cost per megawatt or cost of capacity
 - Genessee 3 cost approximately \$1.5 million/MW or \$1.50 per watt to build
 - Solar panel prices have declined to now lie under \$1/W of capacity
 - Balance of system costs imply that a solar system costs \$2-3/W of installed capacity
- Other prices matter for electricity markets as well
 - Renewable energy credits (usually prices in \$/MWh)
 - Emissions credits or permits (\$/tonne)
 - Capacity payments (\$/MW)
 - GHG or other emissions permits or credits (\$/tonne)

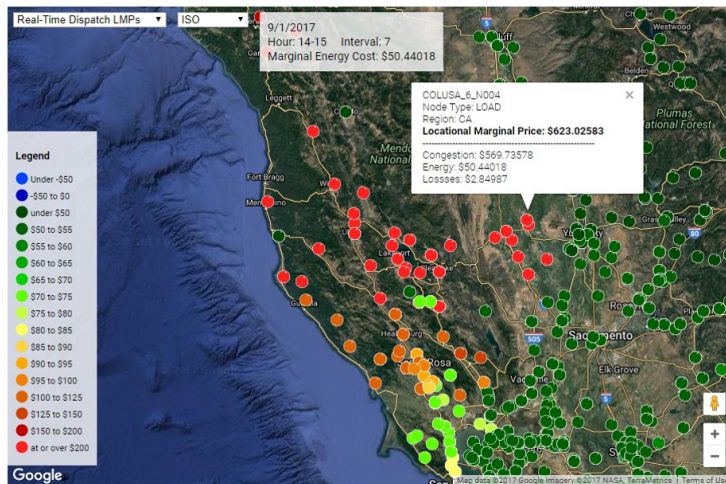
Regulatory characteristics

- Rate-regulated or or state-owned utilities
 - EPCOR (legacy) or BC Hydro
 - PG & E in California
- Competitive markets
 - Energy only markets: ERCOT and Alberta
 - Energy and capacity markets: MISO, PJM
 - Real-time vs day-ahead prices: PJM and others have day-ahead market and then a real-time differences market
 - Many other design characteristic differences between restructured or competitive markets

Alberta Wholesale Energy Market Design

- Energy-only market
- Real time, spot pricing, no day-ahead market (for now)
- Single node (for now?)
- Capacity market was contemplated, but not pursued
- Transmission
 - Regulated monopoly
 - Congestion free (no nodal pricing)
 - No transmission rights
 - Paid for (mostly) by load (consumers)
- Ancillary services: separate, competitive market for operating reserves, transmission-must-run, load-shed and black start

Nodal Pricing Example

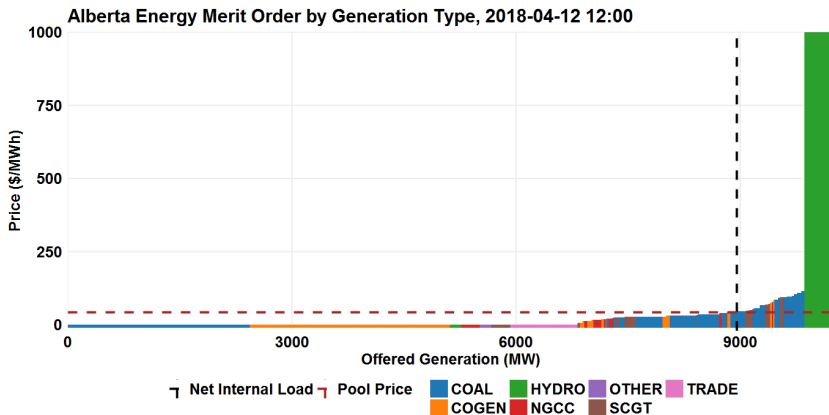


Source: CAISO

The Alberta Wholesale Market

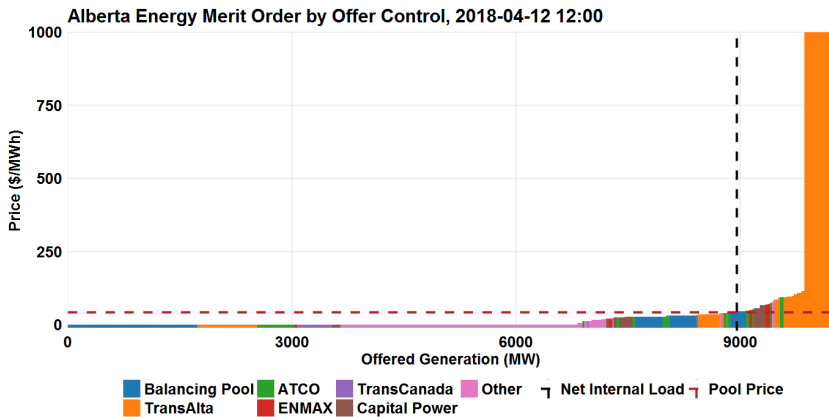
- Suppliers place offers of power at particular price[-5em]
- Demand-side bids placed for power with a maximum price
- Supply offers are sorted from low to high
- Demand offers are sorted from high to low
- Marginal price is set at the price which equates supply and demand - economics 101 at work!
- Import and renewable supply is bid-in at \$0, but everyone receives the market price
- Export demand is bid-in at \$999, so they do not set the price directly but pay the marginal price
- Consumer default bid allows AESO to go up merit order to meet observed demand

The Merit Order



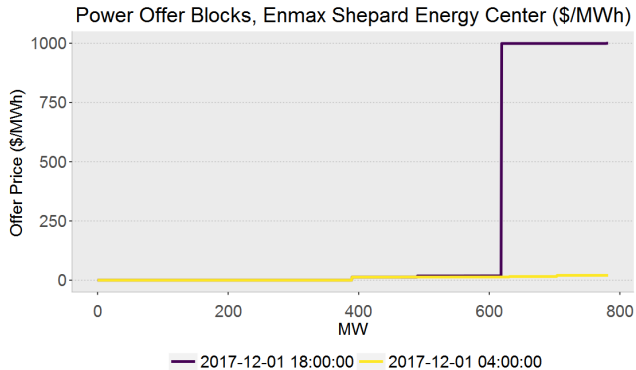
Source: AESO Data, graph by Andrew Leach.

The Merit Order



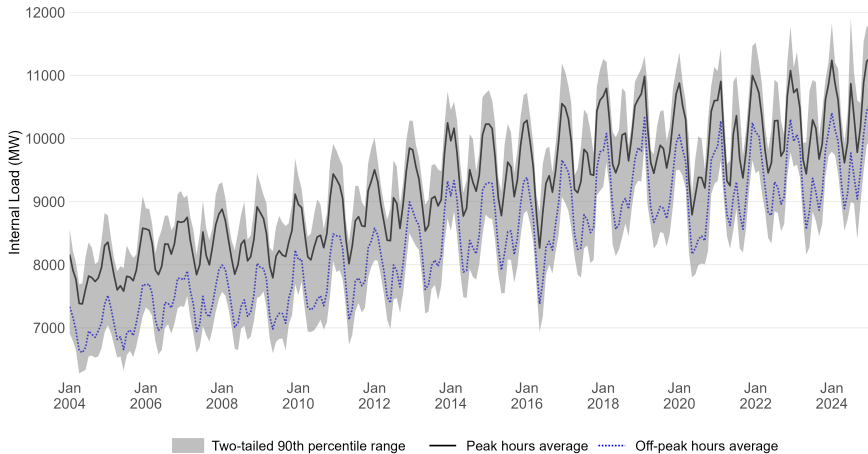
Source: AESO Data, graph by Andrew Leach.

Offers

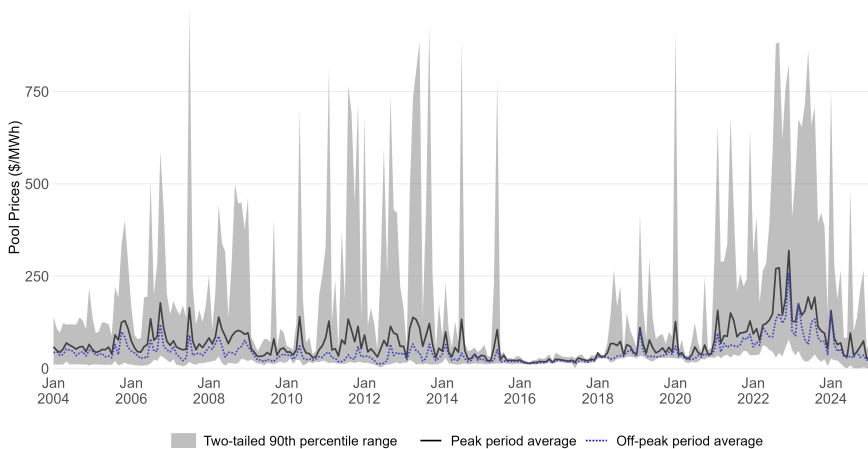


Source: AESO Data, Graph by Andrew Leach

Peak and Average Loads



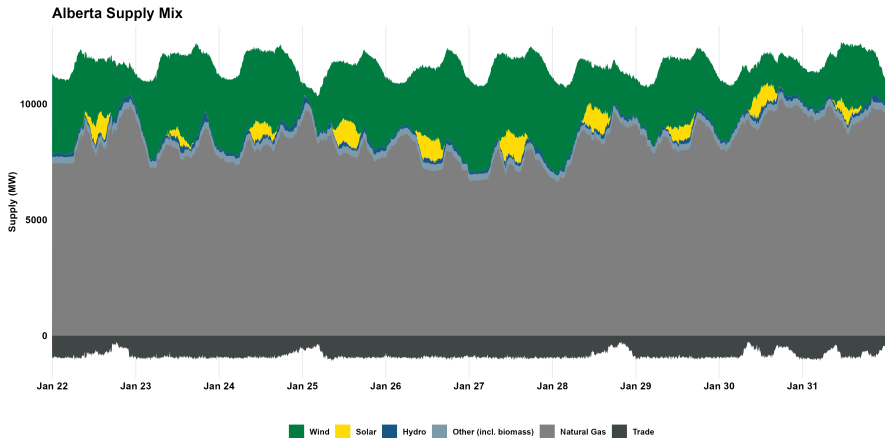
Prices over time



Alberta's Evolving Electricity Market

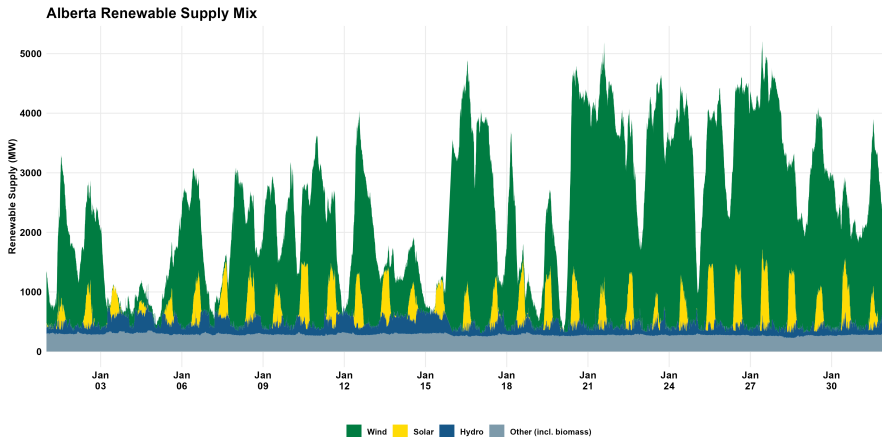
- Capacity Market False Start
- Coal Phase Out
- Renewables (the REP Program and now commercial PPAs)
- Carbon Pricing (TIER and the federal GGPPA)

New Technology



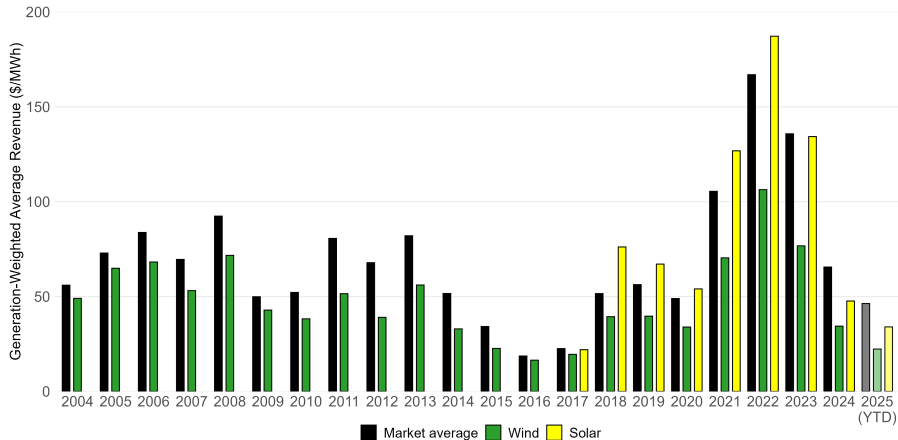
Source: AESO Data, accessed via NRGStream

New Technology



Source: AESO Data, accessed via NRGStream

Price Capture by Technology



Deeper cuts: the roles of storage and transmission

- Renewables now offer some of the cheapest electricity we have ever seen
- Renewables are not (generally) dispatchable
- Renewables seasonal and/or daily generation patterns don't always match load
- The sun is always shining / the wind is always blowing somewhere

Deeper cuts: the roles of storage and transmission

- How do we overcome the need for more transmission or storage?
- Who will pay for the assets? How will the assets be paid for the services they provide?
- Renewables, storage, and even transmission can erode their own value proposition - with a lot of storage or transmission in place, it's harder to see the value for storage
- The value of transmission and storage assets may not be captured by the jurisdiction or the regulatory sector in which they are built